

PADMAPRIYA RAJASEKARAN

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SUMMARY

Manufacturing Engineer with 1.5 years of experience in process development, equipment commissioning, root cause analysis, NPI, and continuous improvement

EDUCATION

Rochester Institute of Technology

Bachelor of Science in Biomedical Engineering, Chemistry Minor

Dec 2026
Dean's List

SKILLS

Manufacturing & Quality: Process Development & Validation, Process Characterization, Continuous Improvement, Root Cause Analysis, DOE, MSA, Gage R&R, FMEA, DFMEA, GD&T, NPI, IQ/OQ/PQ, Work Instructions & SOPs

Software: Minitab, MATLAB, Fusion 360, FEA, Statistical Analysis, Additive Manufacturing

Certifications: Lean Six Sigma Green Belt (CSSC)

WORK EXPERIENCE

Fresenius Medical Care - NPI Systems Engineering Co-op

Jul 2025 - Dec 2025

- Led root cause investigation and developed an 8D report for a verification test setup, implementing corrective actions that reduced measurement error to **0.6%**
- Improved an existing verification procedure by standardizing hardware and optimizing sensor placement, increasing test confidence beyond **95%**
- Established specifications and control limits for **7 process parameters**, including monitoring triggers to identify process deviations
- Developed and executed a SAT protocol for a newly installed reliability test station and performed MSA
- Conducted DOE trials across multiple parameter combinations to validate sensor performance
- Developed a MATLAB-based data pipeline to collect real-time test data to generate machine health reports
- Performed Monte Carlo simulations on test data to establish optimized manufacturing process parameters
- Performed fault injection testing to verify system safe operation limits and documented the test findings
- Tested and approved unreleased hardware for backward compatibility, with three different industry-standard hardware attachments
- Documented process deviations following Good Documentation Practices and collaborated with cross-functional teams to support timely disposition

Cranial Technologies - R&D Engineering Co-op

Jul 2023 - May 2024

- Optimized 3D printer build volume, increasing throughput by **53%** while maximizing material utilization
- Worked with a cross functional organization to transition the manufacturing process of infant cranial helmets from CNC milling to 3D printing, reduced cost by **17%**
- Commissioned an HP Multi Jet Fusion 3D printer for production, collaborating with HP engineers to troubleshoot and hand off for full floor integration
- Designed and built custom jigs, fixtures using GD&T principles to ensure dimensional accuracy of 3 part assembly
- Authored **9 manufacturing WI** and trained **3 operators** on standardized 3D printing and vapor deposition procedures
- Developed and validated quality inspection protocols covering **15 defect types across four product lines**
- Automated 3D scan data conversion by developing a MATLAB tool to process raw 3D scan data into 3D printable STL data, reducing preprocessing time
- Performed destructive testing to analyze mechanical and functional durability of component and benchmarked data used to demonstrate substantial equivalence in the device's FDA submission

PROJECTS

Contact Lens Dry Processing: Redesigned the workflow to integrate two process variables, boosting yield by 19% and eliminating stand-down time

Intravenous Glucose Monitoring System: Benchmarked sensor costs (\$30–\$350) across 16 different options and contributed to the decision matrix. Redefined the problem statement for continuous glucose monitoring and developed an FMEA for the device

Button Controlled Key Turner: Designed and manufactured one handed key turning device and down selected raw material based on safety factor from FEA analysis. Defined must-have design criteria and conducted stakeholder analysis across OTs, patients, insurers, and regulators, translating clinical and market needs into a testable device specification

Physiotherapy Robot: Modeled the kinematics of a 6-DOF therapy robot using Denavit-Hartenberg parameters for a physical therapy application